Prachurjya Sharma

Physical Design Engineer with 3 Years of Experience

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Professional Summary

Physical Design Engineer with hands-on experience in advanced technology nodes (**6nm, 7nm, 12nm and 22nm**). Skilled in the complete **RTL to GDSII** flow with specialization in **Floorplanning**, **Placement**, **Clock Tree Synthesis (CTS)**, **Routing**, and **Sign-off** processes. Proven expertise in **Static Timing Analysis (STA)**, **EM/IR**, **CLP**, **DRC/LVS**, and cross-domain power planning. Proficient in EDA tools like **Innovus**, **PrimeTime**, **Tweaker**, **Aprisa**, and **Calibre**. Adept in scripting with **TCL** and **Perl**. Strong analytical, problem-solving and time management abilities with excellent teamwork, ownership mindset.

Technical Skills

- **Physical Design:** Floorplanning, Placement, CTS, Routing, Timing Closure, Sign-off, ECO
- **Timing/Verification:** STA, Crosstalk, SI, Glitch, EM/IR , CLP, ERC, DRC, LVS, Antenna Checks
- **EDA Tools:** Cadence Innovus, Mentor Aprisa, PrimeTime, Calibre, Tweaker, OpenLane, Magic, Yosys, Opentimer, iverilog, gtkwave
- **Scripting Languages:** TCL, Perl, Verilog, C#, Shell
- **Design Methodologies:** Low Power Design, IR-Aware Design, RC-Aware Routing, Multi Power & Voltage Domain, Metal ECO, Mesh CTS, Area Recovery
- **Soft Skills:** Project Ownership, Team Collaboration, Problem Solving, Deadline Management

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Professional Experience

Quest Global *Bangalore, India*
Physical Design Engineer **June 2022 – Present**

- Contributed to physical implementation of SoC projects across 6nm, 7nm, 12nm (Block Level) and 22nm (Full-Chip) technologies.
- Executed complete block-level flow: Floorplanning, Placement, CTS, Routing, and Sign-off.
- Closed timing, CLP and resolved EM/IR and DRC/LVS violations under tight delivery timelines.
- Automated repetitive DRC using TCL to enhance design productivity.

Projects

Network Chip (MediaTek) **Technology: 6nm** **Jan 2025 – Present**
Tools: Innovus, Tweaker, Calibre **Instances:** 3.4M **Macros:** 77 **Frequency:** 500 MHz

Responsibilities:
- Execution of full flow from Floorplan to Sign-off including Multi-Power Domain Floorplanning and Placement, STA, EM/IR, DRC/LVS.

Challenges:
- Overcame critical placement congestion in multi-power domains with optimized cell density methods.
- Critical timing fixes and metal layer usage for efficient routing for low DRC occurrence.
- Reduced metal DRCs by 25% through custom routing constraints.
- Dynamic and PSH IR drop fixes to reduce resistance by almost 40% and domain crossing CLP fixes.

Wi-Fi Chip (MediaTek) **Technology: 7nm** **Aug 2024 – Jan 2025**
Tools: Innovus, Tweaker, Calibre **Instances:** 3M **Macros:** 87 **Frequency: 1.25 GHz**

Responsibilities:
- Delivered full flow from Floorplan to Sign-off with focus on Mesh CTS, critical setup timing closure, and layout DRC/ IR fixes.

Challenges:
- Latency fixes for hold timing. High metal DRC and LVS fixes with TCL automated scripts.

Television Chip (MediaTek) **Technology: 12nm** **Nov 2023 – Jul 2024**
Tools: Innovus, Tweaker, Calibre **Instances:** 1.68M **Macros:** 113 **Frequency: 206 MHz**

Responsibilities:
- Implemented complete Floorplan to Routing with STA timing fixes for AON partition.
- Delivered multiple Functional ECOs and Metal ECO fixes for hold timing.

Challenges:
- Cell density, DRC and timing criticality in ECO stage along with LEC and CLP fixes for an extreme time-constrained successful tapeout.

Satellite Chip (ORCA Systems) **Technology: 22nm Full-Chip** **Sep 2022 – Aug 2023**
Tools: Aprisa, PrimeTime, Calibre **Instances:** 4M **Macros:** 287 **Frequency: 520 MHz**

Responsibilities:
- Handled Floorplanning (includes die, IO Pad creation and DCAP, level-shifter planning), Power Planning (includes ring and stripe creation) and STA for multi-voltage full-chip design.

Challenges:
- IR Drop analysis and timing fixes across multiple voltage corners.

KPIT Technologies *Pune, India*
Senior Engineer **Feb 2020 – May 2022**

- Developed custom C#/.NET, SQL processes and code for AZURE implementation between cross functional software for Cybersecurity for Automotive ECUs.
- Supported customers across the globe and multi-lingual clients to resolve software issues and understand requirements for implementation.

Certifications & Workshops

– **Advanced Physical Design with OpenLane/SKY130** – GitHub Project

– **TCL for VLSI Design and Synthesis** – GitHub Project

– **RTL Design and Synthesis using Sky130** – GitHub Project

Education

B.Tech. in Electronics and Communication Engineering
SRM Institute of Science and Technology, Chennai **GPA: 81.46%**